

ActuaryGPT — Underwriting Report

Confidential actuarial risk assessment & premium recommendation report.

1. Customer & Policy Profile

Age (Normalized)	0.23	Gender	Male
Height (cm)	175.0	Weight (kg)	70.0
BMI	22.9	Product Info 2	A1
Occupation	WORKER	Annual Income	\$305,000.00
Insurance Type	Life	Coverage Amount	\$4,500,000.00

2. Risk Analysis & Premium Pricing

Predicted Risk Class	Class 1 of 8
Risk Category	Low Risk
Recommended Annual Premium	\$10,000.00

3. AI Actuarial Analysis

Actuarial Report: Life Insurance Application Evaluation

Application Details:

* **Applicant Age:** 0.23 years

* **Height:** 175.0 cm, **Weight:** 70.0 kg, **BMI:** 22.9

* **Smoker:** Yes (1)

* **Occupation:** WORKER, **Income:** \$305,000.00

* **Coverage Amount:** \$4,500,000.00 (Life Insurance)

* **Predicted Risk Class:** 1 (Confidence: 60.1%)

* **Calculated Annual Premium:** \$10,000.00

1. Risk Assessment

The applicant presents a highly contradictory risk profile:

* **Age:** The stated age of 0.23 years indicates an infant. This is the most critical and concerning data point.
* **Physical Attributes (Height, Weight, BMI):** Height of 175 cm and weight of 70 kg result in a BMI of 22.9, which is consistent with a healthy adult male. These attributes are entirely incongruent with an age of 0.23 years.

* **Smoking Status:** "Smoker: 1" is a significant life insurance risk factor and is impossible for an infant.

* **Occupation & Income:** An occupation of 'WORKER' and an income of \$305,000.00 are utterly inconsistent with an age of 0.23 years.

* **Coverage Amount:** A coverage of \$4,500,000.00 is substantial and raises significant questions of insurable interest and anti-selection, especially if the applicant is truly an infant.

* **Previous Claims & Family History:** Both are recorded as 0, which are positive indicators in isolation but are overshadowed by the fundamental data inconsistencies.

In summary, the core applicant data points are in severe conflict, rendering a reliable risk assessment impossible without clarification.

2. Comparative Analysis

The machine learning classifier predicted a Risk Class 1 for the current applicant, leading to a calculated premium of \$10,000.00. This places the applicant in a theoretically lower risk category than the top similar historical approved cases:

* **Current Applicant:** Risk Class 1, Premium \$10,000.00

* **Historical Cases:** Risk Class 2-3, Premiums \$12,000.00 - \$15,000.00

The stated "100.0% Similarity" to these historical cases is highly questionable given the severe internal inconsistencies of the current application (age vs. physical attributes, occupation, income, smoking status). It is improbable that any truly comparable historical case would possess such fundamental data contradictions. The ML model's Class 1 prediction likely reflects the healthy BMI and lack of prior claims/family history, potentially overriding or misinterpreting the anomalous age and smoking status, especially if the features were not robustly

engineered to handle such extreme inconsistencies.

3. Anomaly & Fraud Detection

This application presents severe anomalies:

- * **Extreme Data Inconsistency:** The most prominent anomaly is the profound discrepancy between the applicant's stated age (0.23 years) and all other adult-like demographic and lifestyle attributes (height, weight, BMI, occupation, income, smoking status). This is a critical red flag indicative of either significant data entry error, a system parsing issue, or potential deliberate misrepresentation/fraud.
- * **Impossible Smoker Status:** An infant cannot be a smoker.
- * **High Coverage for Unverified Profile:** The \$4,500,000.00 coverage amount is substantial for any applicant, but particularly problematic for an unverified or potentially infant applicant, raising serious concerns regarding insurable interest and anti-selection.
- * **ML Confidence:** The prediction confidence of 60.1% is moderate, suggesting that even the model exhibited some uncertainty, which is appropriate given the conflicting input.

4. Premium Recommendation

The calculated annual premium of \$10,000.00 corresponds to the ML model's prediction of Risk Class 1. However, this premium recommendation cannot be justified or validated.

The severe and contradictory nature of the input data makes any automated premium calculation unreliable. The ML model's output is compromised by the input data quality. A definitive premium can only be determined once the applicant's true and consistent profile is established through thorough verification. If the applicant is indeed an adult smoker, the premium might require significant adjustment upwards, potentially placing them in a higher risk class than predicted, especially when compared to the similar historical cases.

5. Underwriting Decision Recommendation

Refer for Manual Review (Urgent & Mandatory)

Due to the critical and pervasive data inconsistencies, this application cannot be processed through automated underwriting. Immediate manual review is essential.

Mandatory Actions for Manual Review:

1. **Verify Applicant Identity & Core Data:** Confirm the applicant's true age, height, weight, gender, occupation, income, and smoking status with official documentation.
2. **Investigate Data Source:** Determine the origin of the conflicting data points to understand if it's a data entry error, system glitch, or intentional misrepresentation.
3. **Insurable Interest:** If the applicant is confirmed to be an infant, a thorough investigation into insurable interest for the \$4,500,000.00 coverage amount is paramount.
4. **Re-evaluate Risk Factors:** Once accurate data is obtained, all risk factors must be re-evaluated to determine the appropriate risk class and premium. The current ML model prediction should be entirely disregarded until data integrity is confirmed.

4. Sign-off & Recommendation

Based on the predictive model output and AI underwriting guidelines, the premium recommendation is formally calculated at **\$10,000.00** per annum. The underwriting team should perform secondary manual verification if the risk class exceeds Class 5.